



Department of Computer Science & Information Technology



Course Information

Course Code: **CS-11303**

Credit Hours: 4 **(3,1)**

Lab Hours/Week: **3**

Course Title: **Operating System**

Lecture Hours/Week: **3**

Pre-Requisites: **None**

Teaching Tools and Technologies - LAB

The lab component is centered around practical exposure using Linux-based systems.

Primary Technology Stack

- Ubuntu Linux (LTS version)
- VMware Workstation / VirtualBox
- Linux Command Line Interface (Bash Shell)
- GCC Compiler
- Shell Scripting Environment

Practical Tools Used

1. Process Monitoring Tools

- ps, top, htop
- kill, nice, renice

2. Memory Monitoring Tools

- free, vmstat
- /proc filesystem inspection

3. File System Tools

- mkdir, chmod, chown
- mount, df, du

4. User & Permission Management Tools

- useradd, groupadd
- sudo configuration

5. Shell & Scripting

- Bash scripting
- Cron jobs scheduling

System Programming Tools

- `fork()`, `exec()`, `wait()`
- Pipe and IPC mechanisms
- `pthread`s (basic threading concepts)

Industry Relevance

These tools reflect real Linux administration environments used in:

- Cloud infrastructure
- Cybersecurity operations
- DevOps environments
- Server management

Students gain hands-on competence aligned with entry-level Linux system administrator roles.